Nanoelectronics and Technology

Mid Term Test-2025-Solutions

PART A

Q1. State whether the below statements are True of False **(2 marks)**

F a) MOSFET is a current controlled device

F b) MOSFET shows low resistance path from drain to source under saturation condition

T c) p-channel MOSFET has a p-type doped source and drain

T d) Threshold voltage of a MOSFET is a function of gate dielectric thickness

Q2. What is the main purpose of doping intrinsic silicon? **(1 mark)**

a) To increase its atomic mass
b) To improve its optical transparency
✓ c) To increase its electrical conductivity
d) To change its crystal structure

Q3. When a trivalent impurity is added to silicon, it creates: **(1 mark)**

a) Excess electrons in the conduction band
✓ b) Holes in the valence band
c) No change in carrier concentration
d) Both electrons and holes equally

Q4. What type of impurity is added to create n-type silicon? **(1 mark)**

a) Trivalent impurity
✓ b) Pentavalent impurity
c) Tetravalent impurity
d) Hexavalent impurity

Q5. The concentration of intrinsic carriers (ni) in silicon depends strongly on: **(2 mark)**

a) Pressure
✓ b) Temperature
c) Crystal orientation
d) Impurity type

Q6. What happens to the Fermi level when silicon is doped with a pentavalent impurity?

✓ a) It moves closer to the conduction band **(1 mark)**
b) It moves closer to the valence band
c) It stays at the mid-gap
d) It disappears

Q7. State whether the below statements are True of False **(2 marks)**

T a) Implant energy in ion implantation primarily determines the penetration depth of ions.

F b) Better control over dopant depth and dose is an advantage of diffusion over ion implantation.

F c) During diffusion, the driving force for dopant movement is electric field.

T d) Diffusion less preferred in modern CMOS fabrication compared to ion implantation due to Poor dopant uniformity and depth control.

Q8. State whether the below statements are True of False **(2 marks)**

T a) Film thickness and step coverage in Physical vapour deposition (PVD) are generally *non-uniform* on high-aspect-ratio features.

F b) PVD involves a *chemical reaction* between precursor gases to form a film.

T c) In Chemical vapour deposition (CVD), higher temperatures generally increase reaction rate and film growth rate.

F d) ALD has higher deposition rate compared to CVD.

Q9. State whether the below statements are True of False **(2.5 marks)**

F a) Wet etching generally provides *excellent anisotropy* and precise pattern control.

T b) Wet etching can cause undercutting beneath the mask layer due to its isotropic nature.

T c) Dry etching is generally anisotropic, giving better directionality and feature definition.

T d) Dry etching is preferred for submicron and nanometer-scale pattern transfer.

F e) In reactive ion etching (RIE), only chemical reactions contribute to material removal.

Q10. What causes the hot carrier effect in MOSFETs? **(1 mark)**

a) Excessive gate voltage
b) High electric field near the drain region
c) Low threshold voltage
d) Poor channel mobility

Q11. In a CMOS inverter, noise margin represents: **(1 mark)**

a) The gain of the inverter at the switching point
b) The maximum noise voltage that can be tolerated without logic error
c) The difference between input and output voltages
d) The threshold voltage of the MOSFET

Q12. Which of the following statement/Statements is/are correct with respect to e-beam patterning. **(2 marks)**

F a) Can be used for mass/volume fabrication

T b) Masks are not required

T c) Slow process

T d) High resolution customised pattern with critical dimension-50nm

Q13. Which of the following materials is commonly used for interconnects in modern ICs?

a) Silicon
b) Copper
c) Aluminum
d) Both b and c **(1 Mark)**

Q14. Low-k dielectrics are used in interconnect layers to:
a) Reduce leakage current
b) Reduce parasitic capacitance
c) Increase power dissipation
d) Improve mechanical strength **(1 Mark)**

Q15. As the frequency of an AC signal increases, the skin depth:
a) Increases
b) Decreases
c) Remains constant
d) Becomes independent of material properties **(1 Mark)**

Q16

~~Q12~~. The skin effect occurs due to:
a) Resistance decreasing with frequency
b) Induced eddy currents opposing deeper penetration
c) Increase in the permittivity of the material
d) Thermal effects at high frequency **(1 Mark)**

Q17

~~Q13~~. State whether the below statements are True of False **(1.5 Marks)**

T a) Using a high-k dielectric allows a physically thicker gate insulator while maintaining high capacitance.

F b) High-k materials always have a higher bandgap than SiO_2.

F c) The use of high-k dielectrics increases the equivalent oxide thickness (EOT).

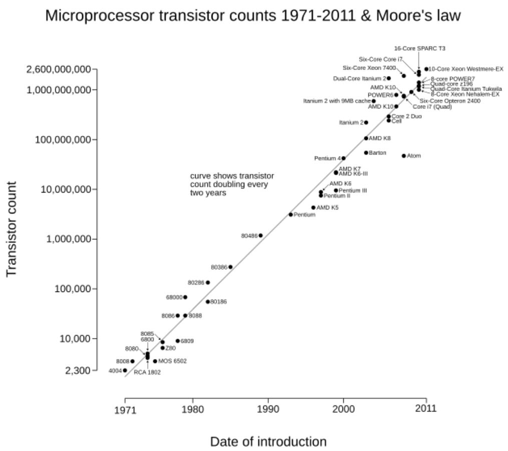
PART B

Q1. Briefly explain the Moore's law in MOSFET Scaling and highlight the key advantages of MOSFET scaling in silicon industry **(3 marks)**

Ans:

Moore's Law is the observation that the number of transistors on an integrated circuit will double every two years with minimal rise in cost. Intel co-founder Gordon Moore predicted a doubling of transistors every year for the next 10 years in his original paper published in 1965.

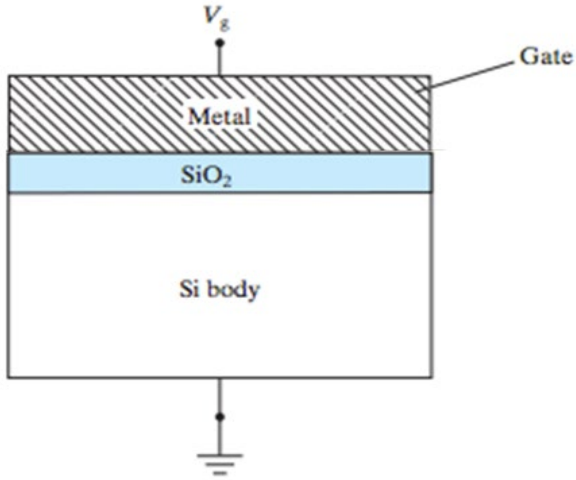
His prediction has proven to be uncannily accurate, in part because the law is now used in the semiconductor industry to guide long-term planning and to set targets for research and development. Moore's Law has been a crucial driver for technological advancement, economic growth, and the development of accessible and innovative products for the global population.



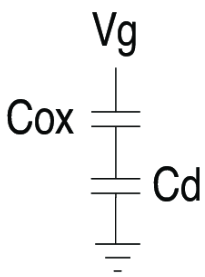
Key advantages:

- Increased Computing Power
- Decreasing Costs
- Smaller and More Efficient Devices
- Innovation in Software and Applications
- Energy Efficiency Improvements
- Economic Growth and Productivity
- Empowerment of New Technologies
- Advances in Scientific Research

Q2) Ignoring the capacitance of the Silicon substrate, write the expression for total capacitance (Ignore capacitance of the Silicon substrate) **(2 marks)**

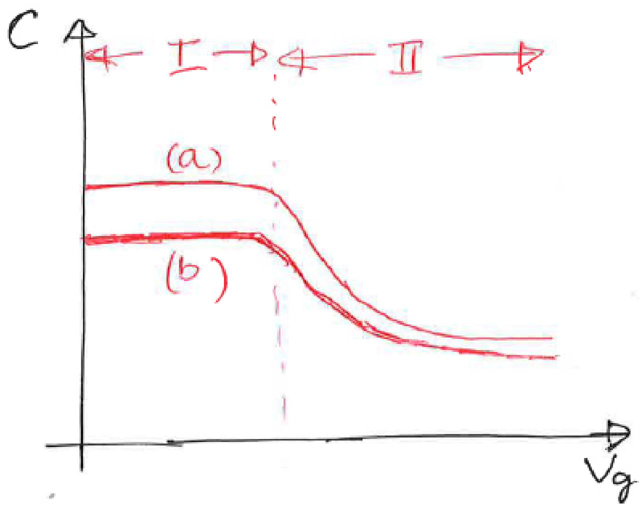


Ans:



$$C_{total} = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_d}}$$

Q3. Consider C-V graph (high frequency) of two MOS capacitors [(a) and (b)] as shown in figure below. Explain the reason for the shift in graph (b) compared to graph (a) only for the section (I). **(2 marks)**

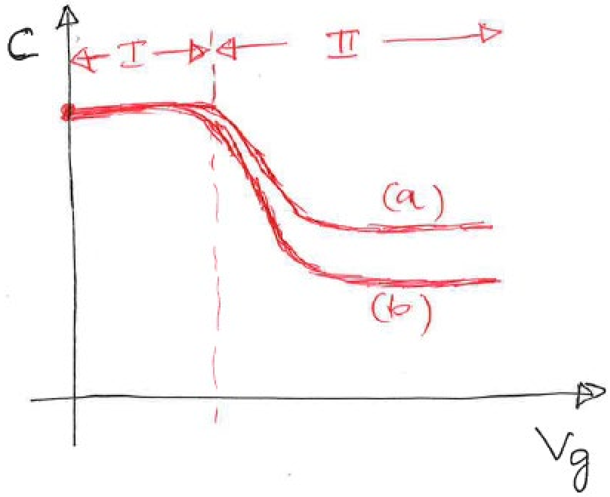


Ans: Considering same dielectric material (same dielectric constant), the shift in graph (b) compared to graph (a) in the section (I) is due to dielectric thickness.

Device (a) has smaller dielectric thickness compared to device (b) dielectric thickness

It can be also compared with respect dielectric constant. (a) has higher K value dielectric compared to (b) with same thickness.

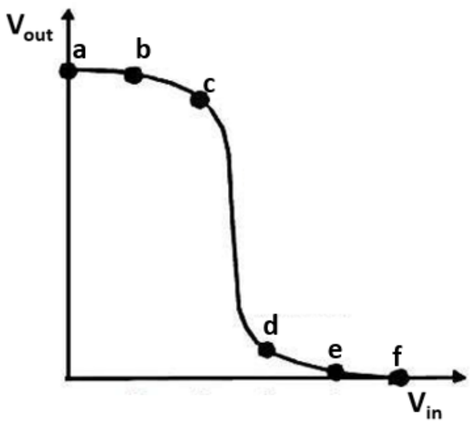
(b) Consider C-V graph (high frequency) of two MOS capacitors [(a) and (b)] as shown in figure below. Explain the reason for the shift in graph (b) compared to graph (a) only for the section (II). **(2 marks)**



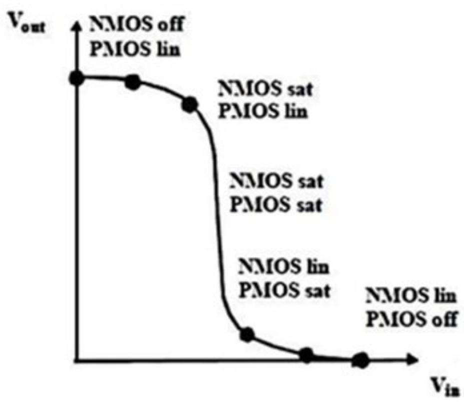
Ans: Considering same dielectric material (same dielectric constant) in both devices (a) and (b), the shift in graph (b) compared to graph (a) in the section (II) is due to silicon substrate doping.

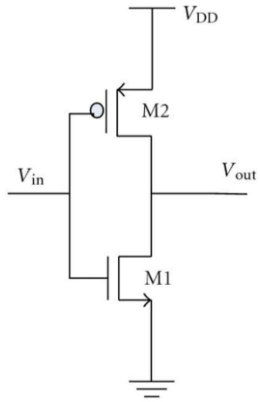
Device (a) has higher substrate doping compared to device (b).

Q4. Draw the schematic of CMOS inverter with appropriate labels. Using the below transfer characteristics curve, indicate the different operating regions of NMOS and PMOS on the curve. **(3 marks)**



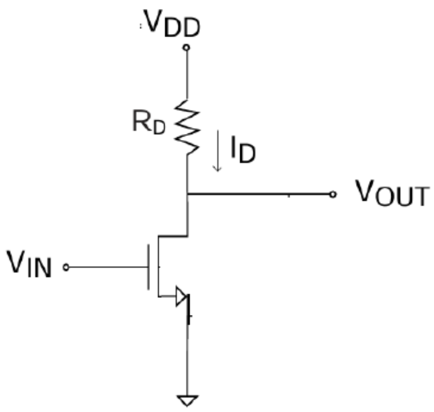
Ans:

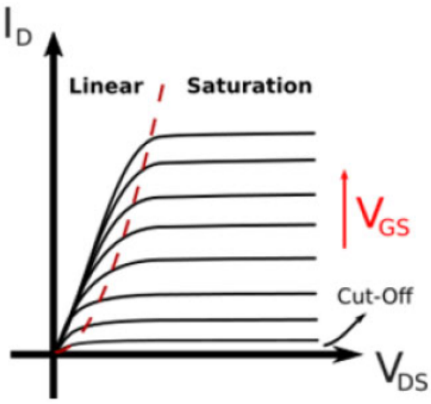




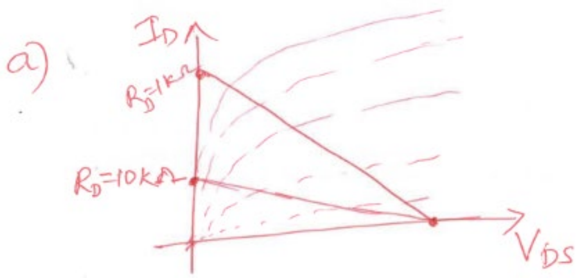
Q5 (a) Draw the DC loadline on the Id-Vds characteristics graph for the below 1T+1R NMOS inverter for R_D=1KΩ and R_D =10KΩ. Ideal NMOS with Vdd=5V, Vin=Vgs and Vout=Vds

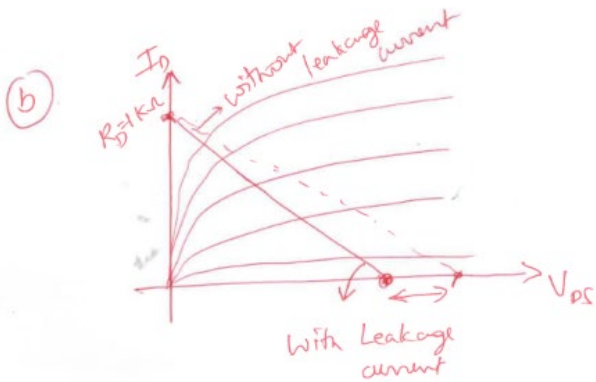
(b) Assume the NMOS has a leakage current of 0.5mA with R_D =1KΩ when Vgs=0. How does the DC loadline change compared to case (a) for R_D =1KΩ. **(4 Marks)**





Ans:





Q6. Briefly explain the velocity saturation effect in advanced MOSFETS. **(2 marks)**

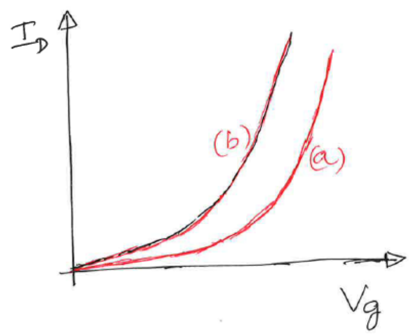
Ans:

Velocity saturation is a short-channel effect in advanced MOSFETs where, at high electric fields, the charge carriers' drift velocity reaches a constant maximum value (saturation velocity) instead of continuing to increase linearly with the electric field. This phenomenon fundamentally alters the device's behavior and performance.

Explanation of the Effect

- **Low Fields:** In long-channel devices or under low electric fields, carrier (electron/hole) velocity is proportional to the electric field strength ($v_d = \mu E$), where μ is the carrier mobility.
- **High Fields:** In advanced, short-channel MOSFETs, the channel length is very small, which results in extremely high electric fields along the channel when a sufficient drain-source voltage (V_{DS}) is applied.
- **The Limit:** When the electric field exceeds a critical value, carriers collide frequently with the crystal lattice (phonon scattering) and lose energy, preventing them from accelerating further. The velocity caps at a material-specific saturation velocity (approximately 1×10^7 cm/s for silicon at room temperature).

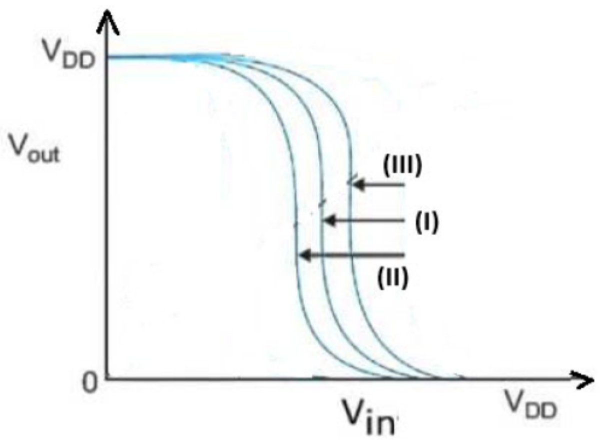
Q7. (a) The I_D-V_g characteristics of two NMOS devices [device 1 and 2] is shown below. The NMOS devices have same doping levels, gate length and width. What parameter of MOSFET responsible for the shift in the graph from [device 1] (a) to [device 2] (b). Highlight the key parameter that can be extracted from the curves (a) and (b) of two respective MOSFETs and compare extracted parameter. **(2 marks)**



Ans: Considering same doping levels, gate length and width in NMOS devices, the shift in graph (b) compared to graph (a) is due to dielectric thickness or dielectric constant. With same dielectric material (same dielectric constant) in NMOS devices (a) and (b), it can be concluded that the shift in the graph is due to dielectric thickness.

Device (b) has thinner dielectric compared to device (b). We can extract the threshold voltages of MOSFETs from the given graphs.

(b) The transfer characteristics curves 3 CMOS inverters is shown below. The curve (I) is obtained from the first inverter with V_{th}(NMOS)=V_{th}(PMOS). Explain the possible reasons for the shift in the curves (II) and (III) of the other two inverters compared to curve (I). **(2 marks)**



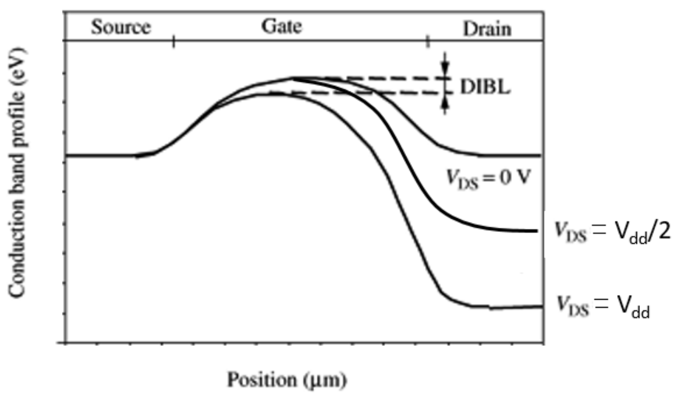
Ans: (II) Stronger NMOS, (III) Stronger PMOS.

Q8. Explain the impact DIBL effect in MOSFET having a gate length L with the help of energy band diagram along the gate length for V_{DS}= 0, (V_{dd}/2), and V_{dd}. **(3 marks)**

Ans:

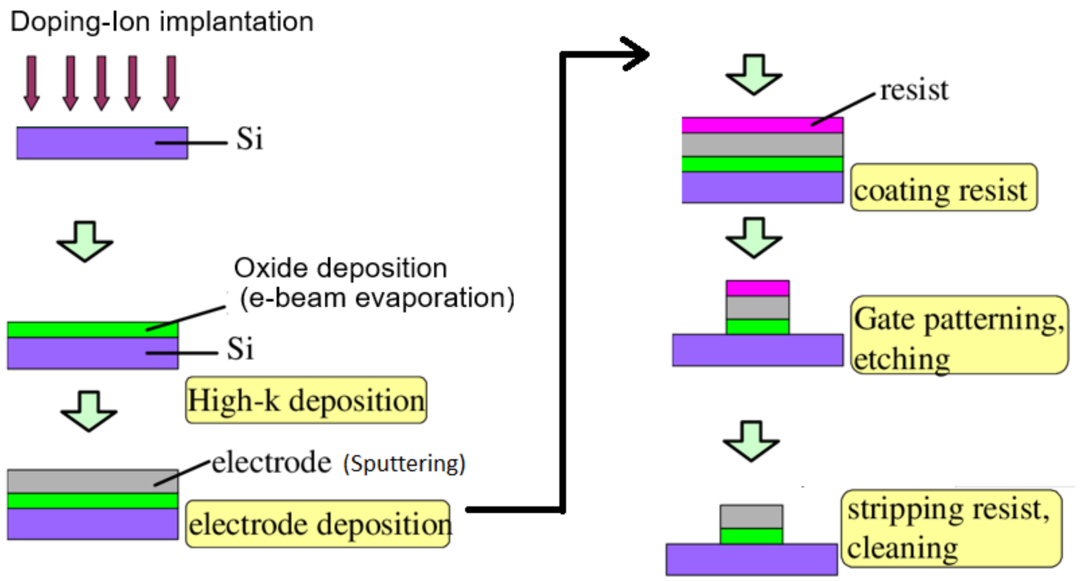
When we apply a positive voltage to the drain of the Short channel device, threshold voltage(V_{th}) will be reduced due to the depletion region under the drain region. Hence, overall channel length will reduce, this is called Drain-induced barrier lowering(DIBL). It is also called as short channel effect in CMOS.

The physical origin of DIBL is the increase of the depletion layer due to a high value of Vds that reduces the equivalent channel length and consequently decreases the threshold voltage. Hence, resulting in increased subthreshold leakage current with a higher drain bias.



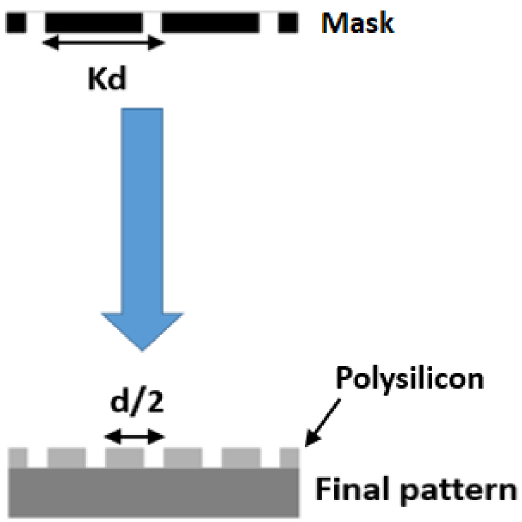
Q9. Clearly present the different steps involved in the fabrication of a MOS capacitor starting from intrinsic silicon substrate, with 10nm thick oxide and Ag as gate electrode. The rectangular area of the capacitor is 50µm x 50 µm. Highlight the specific deposition method used for the deposition of oxide and gate electrode. **(4 marks)**

Ans:

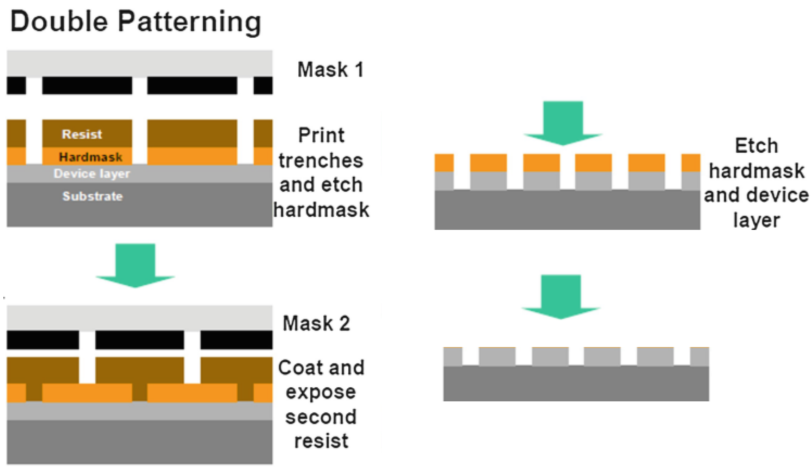


Q10. It is required to fabricate the final polysilicon pattern on Silicon substrate as shown in figure below. Sketch the detailed steps used during the whole lithography process. Optical pitch of the mask is Kd, where K is a constant (>1).

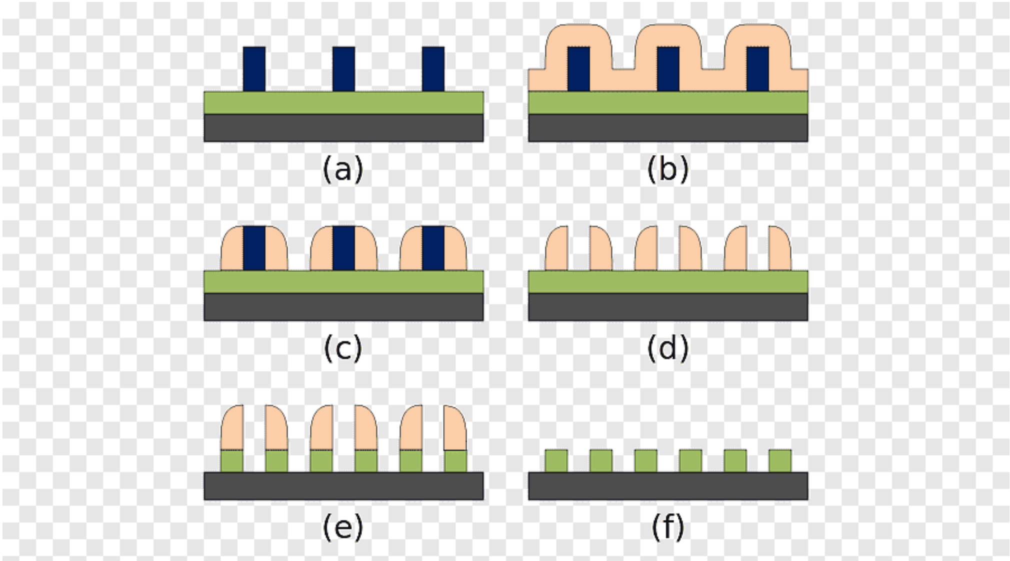
(4 marks)



Ans: Follow double patterning/SADP as shown below



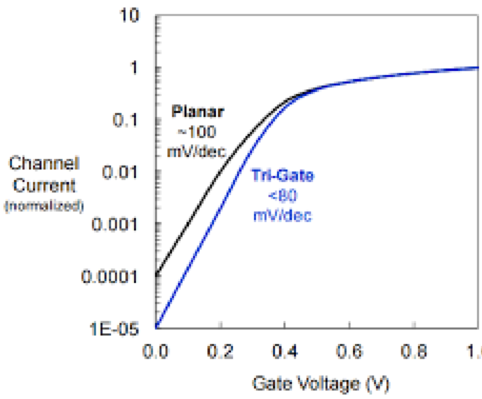
SADP Patterning:



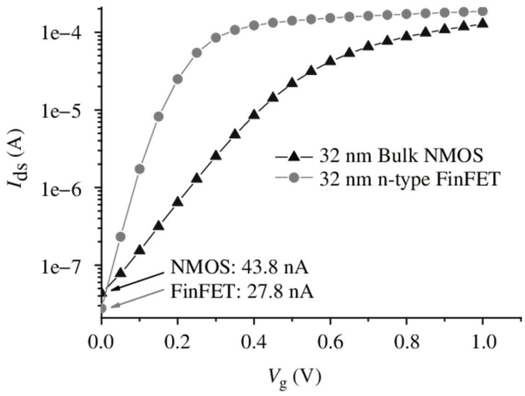
Q11. Discuss the I_{on}, I_{off} and Electric field distribution in FinFETs (Tri-gate) compared to bulk MOSFET. **(3 marks)**

ANS:

- Increased I_{on} due to increased inversion charges occurring from three sides of Fin structure
- Decreased I_{off} due to lower substrate doping
- Non-uniform electric field distribution due to Fin structure resulting in higher electric field at the edges of Fin. Thus, these high electric field regions potentially become dielectric breakdown locations



OFF current



ON current

Q12. Briefly explain the working principle of e-beam sputtering and Atomic Layer Deposition (ALD) technique for the deposition of films/metals. List few applications where e-beam sputtering and ALD are relevant techniques to deposit films/metals. **(4 marks)**

ANS:

E-beam Sputtering (Evaporation)

E-beam evaporation (sputtering is a related but different PVD process) takes place in a high-vacuum chamber where a focused, high-energy electron beam is generated by an electron gun and directed at a target source material.

- The kinetic energy of the electrons is converted into thermal energy upon striking the target, heating it rapidly to its evaporation or sublimation point (often suitable for high-melting-point materials).
- The vaporized atoms then travel in a straight line (line-of-sight process) and condense onto a cooler substrate placed above the source, forming a thin film.
- The process is a physical transport of material, and the film thickness and rate are controlled by the beam power and time.

Atomic Layer Deposition (ALD)

ALD is a gas-phase chemical process based on sequential, self-limiting surface reactions. This approach enables atomic-scale control over film thickness and excellent conformality, even on complex 3D nanostructures. The process typically involves four steps per cycle:

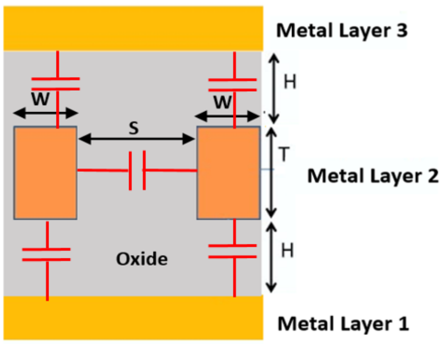
1. **Precursor 1 Exposure**: The first chemical precursor is pulsed into the chamber and adsorbs onto the substrate surface, reacting with available surface groups until all sites are saturated (self-limiting).
2. **Purge**: An inert gas (e.g., argon or nitrogen) is used to purge excess precursor molecules and any reaction by-products from the chamber.
3. **Precursor 2 Exposure**: The second precursor (reactant) is introduced. It reacts with the adsorbed first layer, forming a single atomic layer of the desired film material and generating volatile by-products.
4. **Purge**: Excess second precursor and by-products are purged from the chamber using an inert gas.
   This cycle is repeated until the desired film thickness is achieved, with each cycle adding a precisely controlled layer of material.

Both e-beam sputtering (evaporation) and Atomic Layer Deposition (ALD) are essential thin-film deposition techniques with distinct applications across various industries, particularly in high-tech fields like semiconductors, optics, and aerospace.

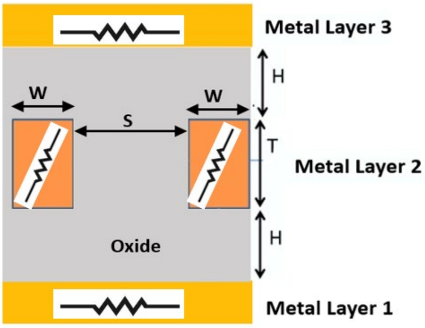
Q13. Indicate Resistance and Capacitance components separately for the interconnect wires shown below. For Wire 1 and 3, resistivity is **ρ_1,** cross sectional area **A_1** and length **L_1**. For Wire 2, resistivity of wire 2 is **ρ_2,** cross sectional area **A_2** and length **L_2**. Dielectric constant of the oxide (dielectric) is ***K*****.** **(3 marks)**

ANS:

Capacitances



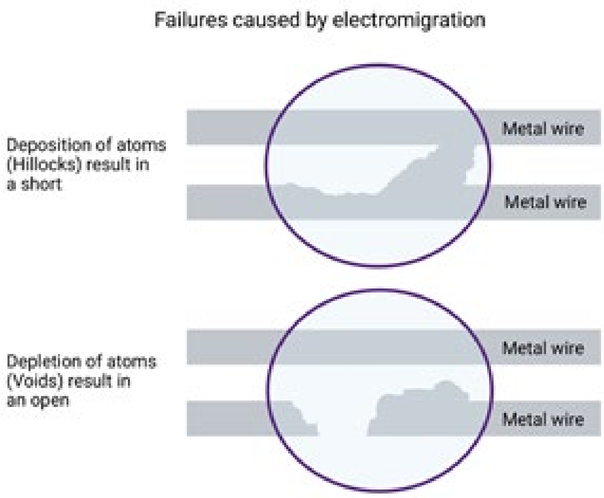
Resistances



Q14. Briefly explain the electro-migration effect in interconnect conducting wires. **(4 marks)**

ANS:

Electromigration is the transport of material caused by the gradual movement of the ions/atoms in a conductor due to the momentum transfer between conducting electrons and diffusing metal atoms. If the current density is high enough, the heat dissipated within the material will repeatedly break atoms from the structure and move them. This will create both 'vacancies' and 'deposits'. The vacancies can grow and eventually break circuit connections resulting in open-circuits, while the deposits can grow and eventually close circuit connections resulting in short-circuit. The effect is important in applications where high direct current densities are used, such as in microelectronic/nanoelectronic devices and related structures.



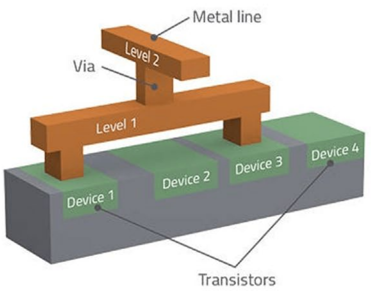
Q15. (a) Why do global interconnects experience higher RC delay compared to local interconnects?

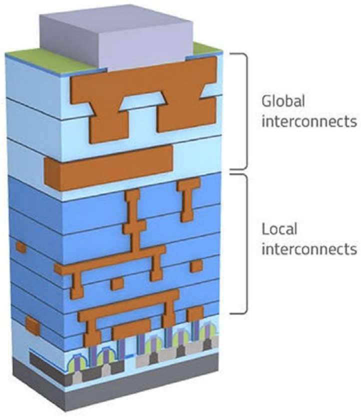
(b) What are the techniques used to reduce RC delay in interconnect technology?

(4 marks)

Ans:

a)





Global interconnects have higher RC (Resistance x Capacitance) delay because they are much longer than local ones, and this increased length directly multiplies the total resistance and capacitance. Additionally, miniaturization in modern chips increases the resistance of the wires and vias, a factor that is compounded over the long distances of global interconnects, further exacerbating the delay problem.

Global interconnects experience higher RC delay because they are **much longer**, have **higher resistance and capacitance**, and RC delay scales with $\mathbf{L^2}$, making long wires significantly slower than short local interconnects.

b)

RC delay is reduced by lowering resistance (R) using copper interconnects, thicker/wider metal lines, and improved barrier/liner materials.
Capacitance (C) is reduced using low-k/ultra-low-k dielectrics, air-gaps, and optimized wire spacing.
Delay is further minimized using repeater insertion and routing global wires on thick top metal layers.